

Domain 1 Study Module: Fundamentals of AI and ML

AWS Certified AI Practitioner (AIF-C01) Exam Preparation

Domain Weight: 20% of scored content (approximately 10 of 50 scored questions) Based on AIF-C01 Exam Guide, Version 1.1 (April 30, 2026) Task 3 of 15 — Study Module

PART 1: DOMAIN INTRODUCTION

What Domain 1 Covers and Why It Matters

Domain 1 is the dictionary and the map for the entire AIF-C01 exam. Every other domain assumes you already know what “model,” “training,” “inference,” and “supervised learning” mean. If Domain 1 is shaky, Domains 2 through 5 will feel like reading a foreign language with no translator.

In plain terms, Domain 1 asks you to do three things:

1. **Speak the language.** Know the core vocabulary of AI and ML well enough to follow a conversation between a data scientist and a business stakeholder.
2. **Recognize the right tool for the job.** Given a business problem, know whether AI/ML is even appropriate, and if so, which technique (classification, regression, clustering) and which AWS managed service fits.
3. **Understand the journey.** Know the stages a model goes through from “we have a business problem” to “the model is live and being monitored in production.”

Domain 1 carries 20% of the exam weight, which translates to roughly 10 of the 50 scored questions. It is not the most heavily weighted domain (that honor belongs to Domain 3), but it is the most foundational. A weak Domain 1 score is rarely caused by Domain 1 being hard — it’s usually caused by rushing past it to get to the “exciting” generative AI content.

What You Must Deeply Understand vs. What You Only Need to Recognize

The exam guide is explicit that this certification is for people who **use** AI/ML solutions, not people who build or code them. That distinction matters for how you study.

Deeply understand (you will be asked to apply this knowledge): - The differences between supervised, unsupervised, and reinforcement learning, and which business problems fit each - The difference between classification, regression, and clustering, and how to pick the right one for a scenario - The eight stages of the ML development lifecycle and what happens at each stage - The difference between batch, real-time, asynchronous, and serverless inferencing, and when each is appropriate - Which AWS managed AI service matches which business use case - The difference between accuracy, precision, recall, and F1 score, and which one matters most in a given scenario (for example, fraud detection)

Only need to recognize (you will not be asked to calculate or implement this): - The mathematical formulas behind precision, recall, or F1 score - How to actually write or tune an algorithm - How to perform hyperparameter tuning yourself - How to build a data pipeline from scratch - The internal architecture of a neural network at a mathematical level

If a study resource asks you to calculate an F1 score by hand or write Python code to train a model, that resource has gone outside the scope of this exam. Recognize the term, recognize what

it measures, and move on.

Five Common Misconceptions Beginners Bring to Domain 1

Misconception 1: “AI and ML are the same thing, just different names.” AI is the broad goal: building systems that can perform tasks that would normally require human intelligence. ML is one specific *approach* to achieving that goal, based on learning patterns from data rather than following hand-written rules. A chess program that follows pre-programmed rules can be “AI” without containing any ML at all. Every ML system is AI, but not every AI system uses ML.

Misconception 2: “Deep learning is something separate from machine learning.” Deep learning is not a rival technology to ML — it is a subset of ML that uses neural networks with many layers to learn complex patterns. Thinking of deep learning as separate from ML is like thinking of a sports car as separate from “cars” — it is a specific, powerful category within the larger category.

Misconception 3: “All AI needs huge amounts of labeled data to work.” This is true for supervised learning, but unsupervised learning works specifically with unlabeled data, finding patterns and groupings without ever being told the “right answer.” Many real business problems — like grouping customers into segments — start with no labels at all.

Misconception 4: “A model that’s 99% accurate is automatically a great model.” Accuracy can be deeply misleading when classes are imbalanced. If 1% of transactions are fraudulent and a model labels every transaction as “not fraud,” it will be 99% accurate while being completely useless. This is exactly why the exam expects you to understand precision, recall, and F1 score as alternatives to accuracy.

Misconception 5: “Reinforcement learning is just supervised learning with extra steps.” Reinforcement learning does not use labeled input-output pairs at all. Instead, an agent takes actions in an environment and learns from rewards and penalties over time, gradually discovering which actions lead to better outcomes. There is no “correct answer” provided up front — the agent has to discover it through trial and error.

Why Domain 1 Is the Foundation for Everything Else

Every later domain leans on Domain 1 vocabulary:

- **Domain 2 (GenAI)** describes large language models and foundation models as a specific, advanced application of deep learning — a term you learn in Domain 1.
- **Domain 3 (FM Applications)** discusses fine-tuning and evaluation metrics that build directly on the training, inference, and evaluation concepts introduced here.
- **Domain 4 (Responsible AI)** revisits bias and variance — terms first defined in Domain 1 — and applies them to fairness and ethics.
- **Domain 5 (Security and Governance)** assumes you understand the ML pipeline (data collection, training, deployment) well enough to recognize where security risks enter that pipeline.

Studying Domain 1 thoroughly now will make every subsequent module faster to learn and easier to retain.

PART 2: OFFICIAL TASK STATEMENT BREAKDOWN

A note on structure: The official AIF-C01 Exam Guide (Version 1.1) defines exactly three task statements for Domain 1: 1.1, 1.2, and 1.3. AWS service identification is not a separate, fourth task statement — it is one of the objectives nested inside Task Statement 1.3 (and referenced again in 1.2). All three official task statements are covered in full below.

Task Statement 1.1: Explain basic AI concepts and terminologies

Full text (from the exam guide): Explain basic AI concepts and terminologies.

Plain-English translation: You need to know what the core words mean — AI, ML, deep learning, neural networks, NLP, computer vision, model, algorithm, training, inferencing, bias, fairness, fit, LLM, GenAI, agentic AI — and how they relate to each other. You also need to know the different “flavors” of inferencing and the different types of data and learning that AI systems use.

Specific learner skills (what you must be able to DO): - *Define* each basic AI term in plain, non-technical language. - *Compare* AI, ML, GenAI, deep learning, and agentic AI, identifying both overlaps and distinctions. - *Distinguish* batch, real-time, asynchronous, and serverless inferencing, and *select* the right type for a given latency or cost requirement. - *Identify* whether a dataset is labeled or unlabeled, and whether it is tabular, time-series, image, text, structured, or unstructured. - *Differentiate* supervised learning, unsupervised learning, and reinforcement learning by the type of data each requires and the type of problem each solves.

Exam watch-out: The single most common mistake is confusing **deep learning** as a competitor to ML rather than a subset of it, and confusing **GenAI** as a totally separate field rather than a specific, advanced application built on deep learning and transformer architectures. If a question asks you to place these terms in a hierarchy, remember: AI is the outermost circle, ML sits inside it, deep learning sits inside ML, and GenAI sits inside deep learning.

Three realistic exam scenario examples (stems only): 1. A company wants to process loan applications overnight in large groups rather than one at a time, since results are not needed immediately. Which type of inferencing should the company use? 2. A retail company has a large dataset of customer purchase histories but no predefined categories for grouping customers. Which type of machine learning is most appropriate? 3. A data scientist describes a system that can write product descriptions, generate marketing images, and answer customer questions in natural language. Which term best describes this system?

Task Statement 1.2: Identify practical use cases for AI

Full text (from the exam guide): Identify practical use cases for AI.

Plain-English translation: Given a business problem, you need to recognize whether AI/ML is a good fit, which specific technique solves it (regression, classification, or clustering), what real-world category the use case falls into (fraud detection, forecasting, computer vision, and so on), and which AWS managed service is built for that exact job. You also need to know when a traditional ML model is the better choice over a foundation model, and vice versa.

Specific learner skills (what you must be able to DO): - *Recognize* business scenarios where

AI/ML adds clear value: supporting human decisions, automating repetitive work, and scaling beyond what a human team could handle. - *Determine* when AI/ML is the wrong tool — for example, when a business needs a guaranteed, deterministic outcome rather than a probabilistic prediction, or when the cost of building and maintaining a model outweighs the benefit. - *Select* the correct ML technique for a stated business need: regression for predicting a number, classification for predicting a category, clustering for discovering groups in unlabeled data. - *Identify* the use case category a scenario belongs to: computer vision, NLP, speech recognition, recommendation systems, fraud detection, forecasting, knowledge bases, or agentic AI. - *Explain* what Amazon SageMaker AI, Amazon Transcribe, Amazon Translate, Amazon Comprehend, Amazon Lex, and Amazon Polly each do. - *Identify* when a traditional ML model is preferable to a foundation model — for example, when strict explainability, low cost, or regulatory transparency is required — and when a foundation model is preferable instead.

Exam watch-out: Learners frequently mix up Amazon Polly and Amazon Transcribe because both deal with audio. Polly turns **text into speech** (think: a robot reading a book aloud). Transcribe turns **speech into text** (think: a court stenographer typing what is said). They are exact opposites of each other, and the exam loves testing this pair.

Three realistic exam scenario examples (stems only): 1. A hospital network needs to predict the likelihood that a specific patient will be readmitted within 30 days, and regulators require the hospital to explain exactly why each prediction was made. Should the hospital use a traditional ML model or a foundation model, and why? 2. A logistics company wants to sort daily shipment volume into “low,” “medium,” and “high” categories to plan staffing. Which ML technique is most appropriate? 3. A media company wants to automatically generate searchable transcripts from thousands of hours of recorded customer service calls. Which AWS service is the best fit?

Task Statement 1.3: Describe the AI/ML development lifecycle

Full text (from the exam guide): Describe the AI/ML development lifecycle.

Plain-English translation: You need to know the stages a model passes through — from collecting data to deploying it and watching it in production — along with where foundation models come from, how models get deployed, which AWS services are used at each stage, the basic ideas behind MLOps, and the performance and business metrics used to judge whether a model is working.

Specific learner skills (what you must be able to DO): - *Describe* and *differentiate* the components of an AI/ML pipeline. - *Describe* where foundation models come from: pre-trained open-source models versus models trained from scratch by an organization. - *Describe* the two main ways to put a model into production: a managed API service (the cloud provider runs the infrastructure) versus a self-hosted API (the customer manages the hosting). - *Identify* the AWS services relevant to each stage of the pipeline, including Amazon Bedrock, Amazon Q, Amazon Quick, Kiro, and SageMaker AI. - *Explain* core MLOps concepts: experimentation, repeatable processes, scalable systems, managing technical debt, production readiness, monitoring, and retraining. - *Describe* model performance metrics (accuracy, precision, recall, F1 score) and business metrics (cost per user, development cost, customer feedback, ROI), and explain what each one tells you.

Exam watch-out: A frequent trap is assuming the ML lifecycle ends at deployment. The exam expects you to know that **monitoring and retraining** are ongoing, permanent stages — a deployed model’s performance can degrade over time as real-world data shifts away from the data it

was trained on (a concept called data drift), and the lifecycle loops back to retraining when that happens.

Three realistic exam scenario examples (stems only): 1. After a model has been in production for six months, a team notices its accuracy has gradually declined even though no code changes were made. What is the most likely cause, and what should the team do next? 2. A company is comparing the cost of hosting its own inference infrastructure on Amazon EC2 versus using a fully managed API. Which deployment method requires the company to manage less infrastructure? 3. A startup wants to evaluate whether its new fraud detection model is actually saving the business money, not just whether the model is statistically accurate. Which category of metric should the startup focus on?

PART 3: CORE CONCEPT NOTES

3A. The AI Hierarchy — What Fits Inside What

Artificial Intelligence (AI): AI is the broad field of building machines that can perform tasks normally requiring human-like judgment — recognizing images, understanding language, making decisions, or solving problems. Think of AI like the category “vehicles.” A vehicle could be a bicycle, a car, a rocket, or a cargo ship — wildly different machines that all share the same broad purpose of getting something from one place to another. AI is the same: an umbrella term covering many different approaches, only some of which involve learning from data.

Machine Learning (ML): ML is the specific approach within AI where a system learns patterns from data instead of being explicitly told the rules. Sticking with the vehicle analogy, ML is like the category “vehicles that move themselves using an engine” — a meaningful subset of all vehicles, but not all of them (a bicycle has no engine, just as some AI has no learning component, such as a simple rules-based chatbot with hard-coded “if-then” responses).

Deep Learning: Deep learning is a subset of ML that uses neural networks with many stacked layers to find complex, layered patterns in data — useful for tasks like recognizing faces in photos or understanding the meaning of a sentence. Continuing the analogy, deep learning is like “engine-powered vehicles with advanced turbocharged engines” — more powerful and capable of handling more complex jobs, but requiring more fuel (data and compute) to run.

Generative AI (GenAI): GenAI is a further specialization within deep learning, built primarily on a type of neural network architecture called a transformer. Instead of just classifying or predicting a single value, GenAI creates new content — text, images, audio, or code — that did not exist before. In the vehicle analogy, GenAI is like a high-end touring vehicle built specifically for one advanced purpose: instead of just moving things from A to B, it can custom-build a unique road as it goes.

Text-based visual — the nested relationship:

ARTIFICIAL INTELLIGENCE (AI)

Broad goal: machines performing human-like tasks

MACHINE LEARNING (ML)

Learns patterns from data instead of fixed rules

DEEP LEARNING

Multi-layer neural networks

GENERATIVE AI (GenAI)

Transformer-based models that
create new content (LLMs, images)

Why this hierarchy matters for the exam: Questions frequently test whether you understand containment. If a question states that “a model uses deep learning,” you can correctly infer it also uses ML and is a form of AI. The reverse is not true — knowing a system “uses AI” tells you nothing about whether ML, deep learning, or GenAI is involved. Watch for trick answers that present these terms as separate, competing categories rather than nested ones.

3B. Types of Machine Learning — A Deep Dive

Supervised Learning

Supervised learning trains a model using labeled data — meaning every example in the training data already has the “correct answer” attached. The model learns the relationship between inputs (called features) and the known outputs (called labels) so that it can predict labels for new, unseen data. Within supervised learning, there are two major problem types: classification, which predicts a category (such as “spam” or “not spam”), and regression, which predicts a continuous number (such as a house price).

A retail bank provides a clear business example: the bank has years of historical loan data, where each past application is labeled with whether the borrower ultimately defaulted or repaid in full. A supervised learning model trained on this labeled history can predict the likelihood that a brand-new applicant will default, helping loan officers make faster, more consistent decisions.

On AWS, supervised learning models are commonly built and trained using **Amazon SageMaker AI**, which provides the full toolkit for training, tuning, and deploying custom supervised models, including labeling tools (SageMaker Ground Truth) to help create the labeled datasets supervised learning depends on.

Unsupervised Learning

Unsupervised learning works with unlabeled data — the model is never told the “right answer.” Instead, it looks for hidden structure, patterns, or groupings within the data on its own. Common tasks include clustering (grouping similar items together), anomaly detection (spotting unusual data points that don’t fit established patterns), and dimensionality reduction (simplifying complex data while preserving its important structure).

A telecommunications company offers a strong business example: the company has millions of customer records with no predefined “customer type” labels. By applying clustering, the company can discover that its customers naturally fall into groups — such as heavy data users, occasional callers, and price-sensitive switchers — without ever having manually defined those groups in advance. This insight then drives targeted marketing campaigns.

On AWS, unsupervised techniques can be built using **Amazon SageMaker AI** (which includes built-in clustering and anomaly-detection algorithms), and unsupervised pattern discovery also underlies parts of **Amazon Personalize**, which detects behavioral patterns to power recommendations.

Semi-Supervised Learning

Semi-supervised learning sits between the other two approaches: it uses a small amount of labeled data combined with a much larger pool of unlabeled data. This matters in the real world because labeling data is often slow and expensive — a human typically has to review and tag each example — while collecting raw, unlabeled data is comparatively cheap and abundant.

A medical imaging company illustrates this well: radiologists can only manually label a few thousand scans as “cancerous” or “healthy” because expert review takes significant time, yet the company has access to millions of unlabeled scans. A semi-supervised approach uses the small labeled set to guide learning while still extracting useful patterns from the much larger unlabeled set, often producing a more capable model than the small labeled set alone could support.

Semi-supervised techniques are typically implemented through custom training pipelines built on **Amazon SageMaker AI**, since this approach requires more customization than a fully managed, single-purpose AI service can offer.

Reinforcement Learning

Reinforcement learning is fundamentally different from the other three types. There is no labeled dataset at all. Instead, an **agent** takes **actions** within an **environment**, receiving **rewards** or **penalties** based on the outcome of each action. Over many attempts, the agent learns a strategy (called a policy) that maximizes its cumulative reward. This makes reinforcement learning especially well suited to problems involving sequential decision-making, where each action affects future options.

A classic, AWS-specific business example is **AWS DeepRacer**, an autonomous racing program where a virtual car (the agent) learns to navigate a track (the environment) by receiving a reward for staying on the track and penalties for going off course. Over thousands of training laps, the car’s policy improves until it can complete the track quickly and reliably. *Exam scope note:* AWS DeepRacer itself does not appear on the official AIF-C01 in-scope or out-of-scope services lists in the current exam guide, so it should be treated as helpful supplemental context for understanding the agent/environment/reward concept, not as a memorized exam-tested service name.

On AWS, reinforcement learning workloads are typically built and trained as custom jobs on **Amazon SageMaker AI**, which supports reinforcement learning frameworks and the specialized training loops this approach requires.

Comparison Table: The Four Types of Machine Learning

Type	Definition	Requires Labels?	Common Use Cases	AWS Connection	Exam Clue Phrase
Supervised Learning	Learns from input-output pairs where the correct answer is already known	Yes, fully labeled data	Fraud classification, price prediction, churn prediction	Amazon SageMaker AI, SageMaker Ground Truth for labeling	“historical data with known outcomes,” “predict a category or number”
Unsupervised Learning	Finds hidden patterns or groupings in data with no known answers	No, fully unlabeled data	Customer segmentation, anomaly detection, market basket analysis	Amazon SageMaker AI (built-in clustering algorithms), Amazon Personalize	“no predefined categories,” “discover groups,” “find patterns”
Semi-Supervised Learning	Combines a small labeled dataset with a much larger unlabeled dataset	Partially, mostly unlabeled	Medical image analysis, speech recognition with limited transcripts	Custom training pipelines on Amazon SageMaker AI	“few labeled examples,” “labeling is expensive,” “mostly unlabeled”
Reinforcement Learning	An agent learns through trial-and-error actions and rewards in an environment	No labels; uses rewards/penalties instead	Robotics, game-playing agents, autonomous navigation	Amazon SageMaker AI (RL training), AWS DeepRacer (conceptual, non-scored)	“agent,” “environment,” “reward,” “trial and error”

3C. The ML Development Lifecycle — All 8 Stages

Stage 1: Business Problem Identification

What happens: Before any data is touched, the organization must define exactly what business problem it is solving and whether that problem genuinely requires ML. Why it matters: A huge percentage of failed ML projects fail not because of bad models, but because the business problem itself was poorly defined or did not need ML in the first place (for example, a problem that needs a guaranteed, deterministic rule rather than a probabilistic prediction). Common challenges: stakeholders disagreeing on success criteria, unclear ROI expectations, and a tendency to reach for ML as a “buzzword solution” when a simpler rules-based system would do the job. AWS tool(s): this stage is largely a business and planning exercise, though AWS Well-Architected guidance and cost-estimation tools like AWS Cost Explorer can inform the decision.

Stage 2: Data Collection and Curation

What happens: Relevant data is gathered from internal systems, third-party sources, or data marketplaces. Why it matters: ML models can only be as good as the data they are trained on — a principle often summarized as “garbage in, garbage out.” Common challenges: data may be incomplete, inconsistent, biased toward certain groups, or simply insufficient in volume. AWS tool(s): **Amazon S3** for storing raw data, **AWS Data Exchange** for licensing third-party datasets, and **AWS Glue** or **AWS Lake Formation** for cataloging and governing collected data.

Stage 3: Data Preparation and Preprocessing

What happens: Raw data is cleaned, normalized, and transformed into a consistent format the model can use — handling missing values, removing duplicates, and converting categorical values into numerical representations (encoding). Why it matters: Inconsistent or messy data directly degrades model accuracy and can introduce hidden bias. Common challenges: deciding how to handle missing data without distorting results, and ensuring preprocessing steps are applied consistently between training and production. AWS tool(s): **AWS Glue**, **AWS Glue DataBrew** (no-code data cleaning), and SageMaker’s built-in data processing capabilities.

Stage 4: Feature Engineering

What happens: Features are the individual measurable inputs a model uses to make predictions (for example, “customer age” or “number of past purchases”). Feature engineering is the process of selecting, transforming, or creating the most useful features from raw data. Why it matters: Well-chosen features often improve model performance more than switching algorithms does. Common challenges: identifying which raw data fields actually carry predictive signal, and avoiding features that leak the answer (for example, accidentally including a future outcome as an input). AWS tool(s): **Amazon SageMaker Feature Store** centralizes and reuses engineered features across teams and projects.

Stage 5: Model Selection and Training

What happens: An algorithm (or a foundation model, in the GenAI context) is chosen and trained on the prepared data, with hyperparameters (configuration settings chosen before training, such as learning rate) adjusted to improve results. Why it matters: Different algorithms suit different problem types and data sizes, and training requires significant compute resources, especially for deep learning. Common challenges: long training times, the cost of compute (especially GPU instances), and choosing between training from scratch versus using a pre-trained model. AWS tool(s): **Amazon SageMaker AI** (training jobs, SageMaker Autopilot for automated model selection, SageMaker JumpStart for pre-built models).

Stage 6: Model Evaluation

What happens: The trained model’s performance is measured against a held-out dataset it has never seen, using metrics like accuracy, precision, recall, and F1 score, and visualized using tools like a confusion matrix (a table showing correct versus incorrect predictions broken down by category). Why it matters: A model that performs well on its training data but poorly on new data has likely **overfit** — memorized the training data rather than learning generalizable patterns. The opposite problem, **underfitting**, occurs when a model is too simple to capture the real pattern in the data at all. Common challenges: choosing the right metric for the business problem (accuracy can be misleading with imbalanced classes), and detecting overfitting before deployment. AWS tool(s): **Amazon SageMaker AI** evaluation and experiment-tracking features.

Stage 7: Model Deployment

What happens: The validated model is made available to receive real input and produce predictions, using either real-time inference (instant predictions for one request at a time, via a hosted endpoint) or batch inference (processing large volumes of data on a schedule, with no immediate response needed). Why it matters: The deployment method must match the business need for speed versus cost-efficiency. Common challenges: choosing between a managed API service (where AWS handles the infrastructure) and a self-hosted API (where the customer manages servers); balancing latency requirements against cost. AWS tool(s): **Amazon SageMaker AI** endpoints for real-time and batch inference, **Amazon Bedrock** for accessing foundation models via a managed API.

Stage 8: Model Monitoring and Maintenance

What happens: Once live, the model's performance and the incoming data are continuously monitored to detect **data drift** (the real-world input data starts looking statistically different from the training data) or **model drift** (the relationship between inputs and the correct outputs changes over time, so even unchanged input data leads to less accurate predictions). When drift is detected, the model is retrained on fresh data. Why it matters: A model that was accurate at launch can silently become inaccurate over months as customer behavior, market conditions, or external factors shift. Common challenges: setting the right thresholds for triggering retraining, and the ongoing cost of monitoring infrastructure. AWS tool(s): **Amazon SageMaker Model Monitor** for automated drift detection and alerting.

3D. AI Use Case Categories — For the Exam

Computer Vision

What it is: AI that interprets and understands visual information from images or video, including image classification (labeling an entire image), object detection (locating and labeling multiple objects within an image), and facial recognition (identifying or verifying individuals). Business examples: a manufacturing plant uses computer vision to automatically spot defective parts on an assembly line before they ship; a retail security team uses it to detect unauthorized access in restricted areas; an insurance company uses it to assess vehicle damage from submitted photos. AWS service: **Amazon Rekognition**. Exam clue phrases: “analyze images,” “detect objects in video,” “identify faces,” “moderate visual content.”

Natural Language Processing (NLP)

What it is: AI focused on understanding, interpreting, and extracting meaning from human language, including sentiment analysis (is this text positive, negative, or neutral?), entity recognition (identifying names, dates, and places within text), and text classification (sorting text into categories). Business examples: a hotel chain analyzes thousands of guest reviews to automatically detect sentiment trends; a law firm extracts key entities like dates and party names from contracts; a customer support team automatically routes incoming tickets to the right department based on content. AWS service: **Amazon Comprehend**. Exam clue phrases: “analyze sentiment,” “extract entities,” “classify text by topic,” “understand the meaning of written content.”

Recommendation Systems

What it is: AI that predicts what a user is likely to want next, based on collaborative filtering (recommending based on what similar users liked) or content-based filtering (recommending based on the attributes of items the user already liked). Business examples: an e-commerce site shows

“customers who bought this also bought” suggestions; a streaming service personalizes its homepage for each viewer; a news app surfaces articles matching a reader’s interests. AWS service: **Amazon Personalize**. Exam clue phrases: “personalized recommendations,” “customers also bought,” “tailor content to each user.”

Fraud and Anomaly Detection

What it is: AI that identifies transactions or behaviors that deviate from established normal patterns, often using unsupervised anomaly detection or supervised classification trained on known fraud examples. Why ML outperforms rule-based systems here: fraud patterns constantly evolve, and fraudsters actively adapt to known rules; ML models can detect subtle, previously unseen patterns that a fixed rule set (for example, “flag any transaction over \$10,000”) would never catch, while a rigid rule set is also prone to flagging large numbers of legitimate transactions as false positives. Business examples: a credit card company flags unusual spending patterns in real time; an insurance company detects suspicious claim patterns; an online marketplace identifies fake account creation at scale. AWS service: custom models built on **Amazon SageMaker AI** (no single fully managed “Fraud Detection” service is on the current in-scope list, so this use case is most often associated with SageMaker AI for the exam).

Predictive Analytics and Forecasting

What it is: AI that predicts future values based on historical patterns, frequently applied to time-series data (data points collected sequentially over time, such as daily sales figures). Business examples: a retailer predicts next quarter’s inventory needs; a utility company forecasts electricity demand by season; a staffing agency predicts seasonal hiring needs. AWS service: forecasting models are typically built using **Amazon SageMaker AI** with its built-in time-series algorithms. *Exam scope note:* Amazon Forecast does not appear on the current AIF-C01 in-scope services list, so do not expect it to be the correct answer choice on this exam even though it conceptually fits this use case.

Speech Recognition and Synthesis

What it is: AI that converts between spoken audio and written text. Automatic speech recognition (ASR) converts speech into text; text-to-speech (TTS) converts text into spoken audio; voicebots combine these with conversational AI to hold spoken dialogues. Business examples: a call center automatically transcribes every customer call for quality review; an airline’s IVR system reads flight status updates aloud to callers; an accessibility tool reads webpage content aloud for visually impaired users. AWS services: **Amazon Transcribe** (speech-to-text), **Amazon Polly** (text-to-speech), and **Amazon Lex** (for building the conversational voicebot logic on top). Exam clue phrases: “convert speech to text” → Transcribe; “convert text to speech” → Polly; “build a voice or chat bot” → Lex.

3E. AWS Services for Domain 1 — In-Scope Deep Dive

Amazon SageMaker AI

One-line purpose: The end-to-end platform for building, training, and deploying custom machine learning models at any scale.

Primary use case: Any scenario requiring a custom model built from your own data, rather than

using an existing pre-built AI service.

Why an AI Practitioner needs to know it: SageMaker AI is the single most heavily referenced service across the entire exam guide and appears in Domains 1, 3, and 4. You won't be asked to use it hands-on, but you must recognize what its major sub-features do.

Key sub-features to know at a recognition level: - **SageMaker Studio**: the integrated development environment (IDE) where data scientists build and manage ML projects. - **SageMaker Canvas**: a no-code, visual interface that lets business analysts build ML models without writing code. - **SageMaker Autopilot**: automatically explores different algorithms and configurations to find a high-performing model with minimal manual effort. - **SageMaker JumpStart**: a curated library of pre-built models and solution templates that can be deployed quickly. - **SageMaker Pipelines**: automates and orchestrates the repeatable steps of an ML workflow (an MLOps capability). - **SageMaker Ground Truth**: helps create high-quality labeled datasets using human labelers and automated labeling assistance. - **SageMaker Feature Store**: a central repository for storing and reusing engineered features across projects. - **SageMaker Model Monitor**: continuously tracks a deployed model's performance and flags data or model drift. - **SageMaker Model Registry**: catalogs and versions trained models to track which version is in production.

Realistic business scenario: A regional bank wanted to build a custom credit risk model trained on its own proprietary loan data, with full control over the algorithm and complete explainability for regulators. They used Amazon SageMaker AI because off-the-shelf AI services could not be trained on their specific dataset or provide the level of customization and audit trail their compliance team required.

Exam clue phrases: "build a custom model," "train your own algorithm," "full control over the ML pipeline," "no-code model building" (Canvas), "automatically find the best model" (Autopilot).

Most commonly confused with: **Amazon Bedrock**. SageMaker AI is for building and training your *own* models from scratch; Bedrock is for accessing and customizing *existing* foundation models through a managed API without training anything from the ground up.

Amazon Rekognition

One-line purpose: A computer vision service that analyzes images and video to detect objects, faces, text, scenes, and inappropriate content.

Primary use case: Any scenario involving understanding the contents of an image or video.

Why an AI Practitioner needs to know it: It is the default exam answer whenever a scenario describes analyzing visual (not textual) content.

Realistic business scenario: A social media platform wanted to automatically detect and remove inappropriate images uploaded by users before they go live. They used Amazon Rekognition's content moderation feature because it can flag explicit or violent imagery in near real time without requiring a human moderator to review every single upload first.

Exam clue phrases: "detect objects in a photo," "identify a face," "moderate image or video content," "analyze video frames."

Most commonly confused with: **Amazon Textract**. Rekognition looks at *what is depicted* in an image (a dog, a car, a face). Textract looks at *what is written* on a document image (the actual printed or handwritten words and form data).

Amazon Comprehend

One-line purpose: A natural language processing service that detects sentiment, entities, key phrases, and topics within text.

Primary use case: Any scenario involving understanding the meaning of existing, already-digitized text.

Why an AI Practitioner needs to know it: It is the default exam answer for any scenario about analyzing the meaning, tone, or topic of written language.

Realistic business scenario: A consumer goods company wanted to understand whether customer sentiment about a recently launched product was trending positive or negative across thousands of support emails. They used Amazon Comprehend because it could automatically classify sentiment at scale without manual review of every message.

Exam clue phrases: “analyze sentiment,” “extract key phrases,” “identify named entities,” “classify documents by topic.”

Most commonly confused with: **Amazon Textract**. Comprehend analyzes text that is already in digital, machine-readable form. Textract is the step that comes *before* Comprehend when the text starts out as an image or scanned document — Textract extracts it, then Comprehend can analyze it.

Amazon Transcribe

One-line purpose: An automatic speech recognition (ASR) service that converts spoken audio into written text.

Primary use case: Any scenario converting recorded or live speech into a text transcript.

Why an AI Practitioner needs to know it: It is the default exam answer for “speech-to-text” scenarios.

Realistic business scenario: A telehealth provider wanted to automatically generate written transcripts of every patient call for medical record-keeping and compliance. They used Amazon Transcribe because it could process recorded calls into accurate text without requiring staff to manually type notes.

Exam clue phrases: “convert speech to text,” “transcribe a recorded call,” “generate captions for video.”

Most commonly confused with: **Amazon Polly**. Transcribe and Polly perform opposite conversions: Transcribe goes from speech to text; Polly goes from text to speech.

Amazon Translate

One-line purpose: A neural machine translation service that converts text between languages.

Primary use case: Localizing content or enabling multilingual communication.

Why an AI Practitioner needs to know it: It is the default exam answer whenever a scenario describes converting content from one *language* to another (as opposed to converting between speech and text, or analyzing sentiment).

Realistic business scenario: An online retailer expanding into new international markets wanted to automatically translate product descriptions and customer reviews into a dozen languages without

hiring a translation team for every market. They used Amazon Translate because it could scale translation across their entire product catalog in real time.

Exam clue phrases: “translate text into another language,” “localize content,” “support multilingual customers.”

Most commonly confused with: **Amazon Comprehend**. Translate changes the language of text; Comprehend analyzes the meaning, sentiment, or entities within text in a single language. They solve entirely different problems despite both being NLP services.

Amazon Polly

One-line purpose: A text-to-speech service that converts written text into natural-sounding spoken audio.

Primary use case: Any scenario requiring written content to be read aloud.

Why an AI Practitioner needs to know it: It is the default exam answer for “text-to-speech” scenarios.

Realistic business scenario: An airline wanted to automatically read flight status and gate change announcements aloud over the airport intercom system without recording a human voice for every possible message combination. They used Amazon Polly because it could generate natural-sounding speech for any text input on demand.

Exam clue phrases: “convert text to speech,” “read content aloud,” “generate natural-sounding audio.”

Most commonly confused with: **Amazon Transcribe**. Again, these are direct opposites — Polly creates speech from text; Transcribe creates text from speech.

Amazon Textract

One-line purpose: A service that extracts printed text, handwriting, and structured data (like form fields and tables) from scanned documents and images.

Primary use case: Digitizing paper documents or scanned PDFs into usable, structured text data.

Why an AI Practitioner needs to know it: It is the default exam answer whenever a scenario describes pulling text or form data *out of an image or scanned document*, as the first step before any further text analysis.

Realistic business scenario: An insurance company received thousands of paper claim forms and wanted to automatically extract the policy number, claimant name, and damage estimate from each scanned form into a database. They used Amazon Textract because it could accurately extract structured form fields without manual data entry.

Exam clue phrases: “extract text from a scanned document,” “pull data from a form or table image,” “digitize paper records.”

Most commonly confused with: **Amazon Rekognition**. Textract reads the *words* on a document image; Rekognition identifies the *visual content* (objects, scenes, faces) within an image. A scanned invoice would use Textract; a photo of a warehouse would use Rekognition.

Amazon Lex

One-line purpose: A service for building conversational interfaces — chatbots and voice bots — powered by the same technology behind Amazon Alexa.

Primary use case: Building a branded, custom conversational experience for customers.

Why an AI Practitioner needs to know it: It is explicitly named in the exam guide’s Task Statement 1.2 as one of six AWS managed AI/ML services you must be able to explain.

Realistic business scenario: A utility company wanted to deploy a chatbot that customers could use to report outages and check restoration estimates through natural conversation, available on both its website and phone system. They used Amazon Lex because it could understand varied, natural phrasing of customer requests and route them appropriately.

Exam clue phrases: “build a custom chatbot,” “voice assistant for customers,” “understand customer intent in conversation.”

Most commonly confused with: **Amazon Polly**. Lex understands and responds to a conversation (speech-to-intent and dialogue management); Polly only converts written text into spoken audio. A bot built in Lex might use Polly behind the scenes to *speak* its responses aloud, but Lex is the “brain” and Polly is the “voice.”

Amazon Personalize

One-line purpose: A service that builds real-time, personalized recommendation systems using machine learning.

Primary use case: Showing each user content, products, or offers tailored to their individual behavior.

Why an AI Practitioner needs to know it: It is the default exam answer for “recommendation engine” scenarios.

Realistic business scenario: A streaming video service wanted to show each user a personalized homepage based on their viewing history rather than the same generic homepage for everyone. They used Amazon Personalize because it could generate individualized recommendations at scale in real time.

Exam clue phrases: “personalized recommendations,” “customers who viewed this also viewed,” “tailor the user experience.”

Most commonly confused with: **Amazon Comprehend**. Personalize works with behavioral and interaction data to generate recommendations; Comprehend works with text content to extract meaning. They are rarely confused in practice, but both are sometimes mistaken for “general purpose AI.”

A brief supplemental note (not on the official in-scope list): AWS DeepRacer is a useful teaching tool for understanding the agent/environment/reward concept behind reinforcement learning (see Part 3B above), and Amazon Forecast is a natural fit for time-series forecasting use cases. Neither currently appears on the official AIF-C01 in-scope services list, so they should be treated as conceptual background rather than memorized, testable services.

PART 4: KEY TERMS AND DEFINITIONS — DOMAIN 1

Term	Simple Definition	Everyday Analogy	Why It Matters for the Exam	Common Misunderstanding to Avoid
Artificial Intelligence (AI)	The broad field of building machines that perform tasks normally requiring human-like judgment.	A toolbox containing every kind of tool, from a simple screwdriver to a power drill.	It is the outermost category; ML, deep learning, and GenAI all sit inside it.	Thinking AI always involves learning from data; some AI is just hard-coded rules.
Machine Learning (ML)	A type of AI where systems learn patterns from data rather than following fixed rules.	A chef who improves recipes by tasting and adjusting over many attempts, rather than following one fixed recipe forever.	It is the foundation term for nearly every other Domain 1 concept.	Treating “AI” and “ML” as interchangeable synonyms.
Deep Learning	A subset of ML using multi-layered neural networks to find complex patterns in data.	An assembly line with many stations, each refining the product a little more before passing it to the next station.	It is the technology underlying most modern computer vision, NLP, and GenAI.	Believing deep learning is unrelated to or separate from ML.
Neural Network	A model structure loosely inspired by layers of connected nodes that process and pass along information.	A relay race where each runner (node) passes a baton (signal) forward, slightly adjusting the pace each time.	You need to recognize the term, not the math behind it.	Assuming you need to understand the underlying calculus to pass the exam; you don’t.
Algorithm	A defined set of steps or rules a computer follows to solve a problem or learn from data.	A recipe: a precise sequence of steps that produces a consistent result.	The exam distinguishes between an algorithm (the method) and a model (the trained result).	Confusing “algorithm” with “model” — they are related but not identical.

Term	Simple Definition	Everyday Analogy	Why It Matters for the Exam	Common Misunderstanding to Avoid
Model (ML)	The trained output produced after an algorithm has learned from data; what makes actual predictions.	A baked cake — the finished product after following the recipe (algorithm) with specific ingredients (data).	The exam tests whether you know a model is the <i>result</i> of training, not the training process itself.	Thinking “model” and “algorithm” mean the same thing.
Training (ML)	The process of feeding data into an algorithm so it can learn patterns and produce a model.	Practicing free throws repeatedly until your shooting form improves.	Training happens before deployment, not during live use.	Confusing training with inference (the two are opposite ends of the model’s life).
Inference	The process of a trained model making a prediction on new, real-world data.	Taking the free-throw shot during an actual game, using the form you practiced.	The exam tests several types of inference (batch, real-time, asynchronous, serverless).	Believing “inference” means the same thing as “training.”
Dataset	A structured collection of data used to train, validate, or test a model.	A textbook full of practice problems used to study for an exam.	Datasets are split into training, validation, and test sets for different purposes.	Assuming one dataset serves all purposes without being split.
Feature	An individual measurable input variable used by a model to make predictions.	One ingredient in a recipe, such as flour or sugar, that contributes to the final result.	Feature engineering (Stage 4 of the lifecycle) is built entirely around this concept.	Confusing a “feature” with a “label” — features are inputs, labels are answers.
Label	The known, correct answer attached to a piece of training data in supervised learning.	The answer key attached to a practice test question.	Labels are what make a dataset “labeled” versus “unlabeled.”	Thinking unsupervised learning also uses labels — it does not.

Term	Simple Definition	Everyday Analogy	Why It Matters for the Exam	Common Misunderstanding to Avoid
Ground Truth	The verified, accurate real-world answer used to label or evaluate training data.	The actual final score of a game, used to check whether a prediction was correct.	SageMaker Ground Truth is named directly after this concept.	Confusing ground truth with a model's prediction — ground truth is the verified fact, not a guess.
Supervised Learning	Learning from data where every example already has a known, correct label.	Studying with flashcards that show both the question and the correct answer.	One of the three (or four, including semi-supervised) core learning types tested directly.	Believing supervised learning can work without any labeled data.
Unsupervised Learning	Learning that finds hidden patterns or groupings in data with no labels provided.	Sorting a junk drawer into logical piles without anyone telling you the categories in advance.	Frequently tested alongside clustering and anomaly detection use cases.	Assuming unsupervised learning has no business value because there are no “right answers.”
Semi-Supervised Learning	Learning that combines a small labeled dataset with a much larger unlabeled dataset.	Learning a new subject from a few worked example problems plus a much larger pile of unworked practice problems.	Tested as the “middle ground” option in scenario questions about limited labeled data.	Thinking semi-supervised learning requires no labeled data at all.
Reinforcement Learning	Learning where an agent takes actions in an environment and learns from rewards and penalties.	Training a dog with treats: good behavior is rewarded, and the dog learns through repeated trial and error.	Tested through agent/environment/vocabulary and scenario recognition.	Believing reinforcement learning uses labeled input-output pairs like supervised learning.

Term	Simple Definition	Everyday Analogy	Why It Matters for the Exam	Common Misunderstanding to Avoid
Overfitting	When a model learns the training data too precisely, including its noise, and performs poorly on new data.	A student who memorizes the exact practice questions and answers but fails when the real exam asks the questions differently.	A core evaluation concept tested in Domain 1 and revisited in Domain 4.	Assuming a model that performs perfectly on training data is automatically a good model.
Underfitting	When a model is too simple to capture the real underlying pattern in the data.	Trying to summarize an entire novel in one short sentence — too much important detail is lost.	The opposite failure mode from overfitting; both are commonly tested together.	Thinking underfitting and overfitting are the same problem.
Bias (Statistical / ML)	A systematic error in a model's predictions, often caused by unrepresentative training data.	A scale that is mis-calibrated and always reads five pounds heavier than the true weight.	Connects Domain 1 directly to fairness concepts in Domain 4.	Confusing statistical bias with general human prejudice — though the two can be related.
Variance	How much a model's predictions change when trained on slightly different data samples.	A dart player whose throws land in wildly different spots each round, even when aiming the same way.	High variance is associated with overfitting.	Believing low variance always means a good model — a consistently wrong model has low variance but is still wrong.
Accuracy	The percentage of all predictions a model gets correct.	A trivia player's overall percentage of questions answered correctly.	Can be misleading on imbalanced datasets — a key exam trap.	Assuming high accuracy always means a model is useful in practice.
Precision	Of all the items a model predicted as positive, the percentage that were actually positive.	Of everyone you accused of cheating, the percentage who actually cheated.	Important when false positives are costly (for example, flagging legitimate transactions as fraud).	Confusing precision with recall — they answer different questions.

Term	Simple Definition	Everyday Analogy	Why It Matters for the Exam	Common Misunderstanding to Avoid
Recall	Of all the items that were actually positive, the percentage the model correctly identified.	Of everyone who actually cheated, the percentage you successfully caught.	Important when false negatives are costly (for example, missing actual fraud or disease).	Believing high precision automatically means high recall, or vice versa.
F1 Score	A single score that balances precision and recall together.	A combined report card grade that accounts for two different subjects equally.	Useful when you need one number to compare models instead of two.	Assuming F1 score is just the average of accuracy and precision — it is a distinct calculation.
Confusion Matrix	A table showing a model's correct and incorrect predictions broken down by actual versus predicted category.	A scorecard showing not just how many shots you made, but exactly which target you were aiming at each time.	Helps visually explain where precision and recall come from.	Thinking a confusion matrix only applies to two-category problems — it works for multiple categories too.
Hyperparameter	A configuration setting chosen before training that controls how the model learns.	The oven temperature and bake time you set before baking, rather than something the cake decides on its own.	The exam expects recognition only, not tuning skills.	Confusing hyperparameters with the patterns the model learns during training (those are called parameters).
Epoch	One complete pass of the entire training dataset through the learning algorithm.	One full read-through of a textbook before starting a second read-through.	A recognition-level vocabulary term.	Believing one epoch means training is complete — multiple epochs are typically needed.

Term	Simple Definition	Everyday Analogy	Why It Matters for the Exam	Common Misunderstanding to Avoid
Batch	A subset of the training data processed together in one step during training, or a group of records processed together during inference.	Grading a stack of ten exams together before moving to the next stack.	Connects directly to “batch inferencing” as a deployment concept.	Confusing a training batch with a batch inference job — related idea, different context.
Validation Set	A portion of data held back from training, used to tune the model and check progress during development.	A practice quiz used to check your understanding before the real exam.	Tested directly in the training/validation/test comparison table.	Confusing the validation set with the test set — validation is used <i>during</i> development, not at the very end.
Test Set	A portion of data the model has never seen at all, used for a final, unbiased performance check.	The real final exam, taken only once, after all studying and practice quizzes are done.	The final, most objective measure of real-world performance.	Using the test set repeatedly during development — this defeats its purpose.
Training Set	The largest portion of data, used to actually teach the model the underlying patterns.	The textbook chapters and practice problems you study from.	Forms the foundation of the model; described in the lifecycle’s training stage.	Assuming the training set alone is enough to evaluate a model fairly.
Regularization	A technique that discourages a model from becoming too complex, helping prevent overfitting.	Setting a strict word limit on an essay to force concise, focused writing instead of rambling.	A recognition-level term connected to overfitting prevention.	Believing regularization always improves accuracy — it trades some training accuracy for better generalization.

Term	Simple Definition	Everyday Analogy	Why It Matters for the Exam	Common Misunderstanding to Avoid
Dropout	A regularization technique (mainly in neural networks) that randomly ignores some nodes during training.	A sports team practicing with random players sitting out each drill, forcing the team to not rely on any single star player.	A recognition-level term you should be able to define if asked.	Assuming dropout removes nodes permanently — it only ignores them temporarily during training.
Activation Function	A mathematical function inside a neural network node that decides whether and how strongly a signal passes forward.	A light switch with a dimmer, deciding how much of a signal to let through rather than a simple on/off.	Recognition-level only; no calculation required.	Thinking you need to know specific activation function formulas for this exam — you do not.
Computer Vision	AI focused on interpreting visual information from images and video.	Teaching a machine to “see” and describe what’s in a picture the way a person would.	Directly tied to Amazon Rekognition use cases.	Confusing computer vision (interpreting images) with image generation (creating new images, a GenAI topic).
Natural Language Processing (NLP)	AI focused on understanding and working with human language.	Teaching a machine to read and understand the meaning of a paragraph, not just recognize letters.	Directly tied to Amazon Comprehend, Translate, Lex use cases.	Assuming NLP only means translation — it covers sentiment, entities, classification, and more.
Speech Recognition	AI that converts spoken audio into text.	A stenographer typing exactly what is being said in a courtroom.	Directly tied to Amazon Transcribe.	Confusing speech recognition with text-to-speech, which is the opposite direction.
Clustering	An unsupervised technique that groups similar data points together without predefined categories.	Sorting a basket of mixed fruit into piles by similarity without being told the fruit names in advance.	A core unsupervised learning technique tested directly.	Believing clustering requires labeled categories to start with — it does not.

Term	Simple Definition	Everyday Analogy	Why It Matters for the Exam	Common Misunderstanding to Avoid
Classification	A supervised technique that assigns data into one of a set of predefined categories.	Sorting mail into “junk” and “important” bins based on learned patterns.	Frequently contrasted with regression in scenario questions.	Confusing classification (predicting a category) with regression (predicting a number).
Regression	A supervised technique that predicts a continuous numerical value.	Estimating a house’s sale price based on its size, location, and condition.	Frequently contrasted with classification in scenario questions.	Assuming regression always produces a percentage — it can predict any continuous number.
Anomaly Detection	A technique (often unsupervised) for identifying data points that deviate significantly from normal patterns.	A security guard who notices one person acting strangely in a crowd of otherwise normal behavior.	Connected directly to fraud detection use cases.	Believing anomaly detection always requires labeled “fraud” examples — it often works without any labels at all.
Data Preprocessing	Cleaning and transforming raw data into a consistent, usable format before training.	Washing and chopping vegetables before you start actually cooking a meal.	Stage 3 of the ML lifecycle.	Skipping this step mentally when thinking through the lifecycle — it’s easy to underestimate its importance.
Feature Engineering	Selecting, transforming, or creating the most useful input variables for a model.	Deciding which ingredients actually improve a recipe’s flavor, and adjusting their proportions.	Stage 4 of the ML lifecycle.	Confusing feature engineering with data preprocessing — preprocessing cleans data; feature engineering shapes which inputs the model actually uses.

Term	Simple Definition	Everyday Analogy	Why It Matters for the Exam	Common Misunderstanding to Avoid
Model Deployment	Making a trained model available to receive input and generate predictions in the real world.	Opening a restaurant to the public after all the recipes have been perfected in test kitchens.	Stage 7 of the ML lifecycle; connects to real-time vs. batch inference.	Thinking deployment is the final stage — monitoring and maintenance still follow.
Data Drift	When the statistical properties of real-world input data shift away from the data the model was originally trained on.	Training for a marathon on flat roads, then being asked to race on a mountain trail — the “terrain” (data) has changed.	A key reason models need ongoing monitoring and retraining.	Confusing data drift (the inputs changing) with model drift (the relationship between inputs and outputs changing).
Model Drift	When a model’s predictive accuracy declines over time because the relationship between inputs and correct outputs has changed.	A weather forecaster whose old climate patterns no longer hold true after the climate itself has shifted.	Tested alongside data drift as a trigger for retraining.	Believing model drift is caused by a coding bug — it’s typically caused by real-world change, not broken code.

PART 5: COMPARISON TABLES — DOMAIN 1

Table 1: AI vs. ML vs. Deep Learning vs. Generative AI

Category	Definition	Key Characteristic	Requires Training Data?	Example	AWS Service Example
AI	The broad field of building machines that perform human-like tasks	Umbrella term covering many techniques, including non-learning approaches	Not always (rules-based AI may need none)	A simple rules-based chatbot that replies to fixed keywords	Amazon Lex (rules + ML combined)

Category	Definition	Key Characteristic	Requires Training Data?	Example	AWS Service Example
ML	Systems that learn patterns from data rather than fixed rules	Improves performance through exposure to data	Yes, always	A model predicting customer churn from account history	Amazon SageMaker AI
Deep Learning	A subset of ML using multi-layered neural networks	Excels at complex, layered pattern recognition	Yes, typically large volumes	A model recognizing objects in photos	Amazon Rekognition (built on deep learning)
Generative AI	A subset of deep learning that creates new content using transformer architectures	Produces novel text, images, audio, or code rather than a fixed prediction	Yes, massive pre-training datasets	A chatbot that writes a custom email draft	Amazon Bedrock

Table 2: Supervised vs. Unsupervised vs. Semi-Supervised vs. Reinforcement Learning

Type	Data Requirement	Goal	Example Output	Common Pitfall
Supervised	Fully labeled data	Predict a known category or number for new data	“This transaction is fraudulent”	Requires costly, time-consuming labeling effort
Unsupervised	Fully unlabeled data	Discover hidden structure or groupings	“These five customer segments exist”	Results can be harder to validate without ground truth
Semi-Supervised	Small labeled set plus large unlabeled set	Improve learning when full labeling is impractical	“This scan is likely cancerous, based on limited examples”	Quality depends heavily on how representative the small labeled set is
Reinforcement	No labels; uses rewards and penalties	Learn an optimal sequence of actions over time	“This is the fastest safe path around the track”	Requires a well-designed reward system, or the agent learns the wrong lesson

Table 3: Training Set vs. Validation Set vs. Test Set

Dataset	Purpose	When Used	What Happens If You Get It Wrong
Training Set	Teaches the model the underlying patterns in the data	During the initial training process	The model has nothing to learn from, or learns from too little data to generalize well
Validation Set	Tunes the model and checks progress before final evaluation	During development, often repeatedly, to compare model versions	The model may be indirectly “tuned” to the validation set, producing overly optimistic results
Test Set	Provides a final, unbiased measure of real-world performance	Only once, after training and tuning are complete	If reused during development, it stops being a fair, unbiased test and performance estimates become unreliable

Table 4: Overfitting vs. Underfitting

Aspect	Overfitting	Underfitting
Definition	Model learns training data too precisely, including noise	Model is too simple to capture the real underlying pattern
Symptoms	Excellent performance on training data, poor performance on new data	Poor performance on both training data and new data
Causes	Overly complex model, too little training data, training for too long	Overly simple model, insufficient features, training stopped too early
Solutions	Add regularization, gather more data, simplify the model, use dropout	Use a more complex model, add more relevant features, train longer

Table 5: Classification vs. Regression vs. Clustering

Technique	What It Predicts	Example	Evaluation Metric
Classification	A category or class label	“Spam” or “Not Spam”	Accuracy, precision, recall, F1 score
Regression	A continuous numerical value	Predicted house price: \$342,000	Metrics based on the size of prediction error (recognition level only)

Technique	What It Predicts	Example	Evaluation Metric
Clustering	Natural groupings within unlabeled data	“These customers form Segment A, B, and C”	Evaluated by how distinct and meaningful the discovered groups are (recognition level only)

Table 6: Real-Time Inference vs. Batch Inference

Aspect	Real-Time Inference	Batch Inference
Latency	Immediate, typically milliseconds to seconds	Delayed; results may take minutes to hours
Cost	Generally higher, since infrastructure must stay ready at all times	Generally lower, since resources run only when the batch job executes
Use Case	Live fraud detection during a transaction; chatbot responses	Generating end-of-month sales reports; bulk document processing overnight
AWS Option	SageMaker AI real-time endpoints	SageMaker AI batch transform jobs

Table 7: Model Accuracy vs. Precision vs. Recall vs. F1 Score

Metric	What It Measures	Best Used When	Limitation
Accuracy	The overall percentage of correct predictions across all categories	Classes are roughly balanced	Misleading when one class is rare (for example, fraud detection)
Precision	Of predicted positives, how many were truly positive	False positives are costly (for example, blocking legitimate transactions)	Says nothing about how many true positives were missed
Recall	Of actual positives, how many were correctly identified	False negatives are costly (for example, missing a fraudulent transaction or a disease diagnosis)	Says nothing about how many false alarms were raised
F1 Score	A balance between precision and recall in a single number	You need one combined score to compare models fairly	Can mask cases where precision and recall are wildly different from one another

PART 6: 25 DOMAIN 1 PRACTICE QUESTIONS

Beginner Questions (1–10)

Question 1 (Beginner) A company describes a software program that follows a fixed set of “if this, then that” rules written by a developer, with no ability to learn or improve from data. Which statement is most accurate?

A. This program cannot be considered AI under any definition. B. This program is a form of AI, but it is not a form of machine learning. C. This program is a form of machine learning, but not deep learning. D. This program is a form of deep learning.

Correct Answer: B

Explanation: AI is a broad category that includes systems that follow fixed, hand-coded rules, since AI describes the goal (performing human-like tasks) rather than the specific method used. Because this program never learns from data, it does not qualify as machine learning, which by definition requires learning patterns from data.

Why the others are wrong: A. Rules-based systems are a long-recognized category of AI; the lack of learning doesn’t disqualify it from being AI. C. ML requires learning from data; a fixed rule set never learns, so it cannot be ML. D. Deep learning is a more specific and advanced subset of ML, which this program is not.

Task statement mapping: Task 1.1 — Explain basic AI concepts and terminologies. Key concept tested: AI vs. ML distinction.

Question 2 (Beginner) Which of the following best describes the relationship between machine learning and deep learning?

A. They are unrelated technologies that solve different types of problems. B. Deep learning is a broader category that contains machine learning. C. Machine learning is a subset of deep learning. D. Deep learning is a subset of machine learning that uses multi-layered neural networks.

Correct Answer: D

Explanation: Deep learning is a specialized subset of machine learning that relies specifically on neural networks with multiple layers to detect complex patterns in data.

Why the others are wrong: A. They are directly related — deep learning is built within the broader ML category, not separate from it. B. This reverses the actual hierarchy; ML is the broader category, not deep learning. C. This also reverses the hierarchy incorrectly.

Task statement mapping: Task 1.1 — Explain basic AI concepts and terminologies. Key concept tested: AI/ML/deep learning hierarchy.

Question 3 (Beginner) A dataset contains thousands of customer support tickets, with each ticket already tagged as “billing,” “technical,” or “account access” by past support staff. What kind of data is this?

A. Unlabeled data B. Labeled data C. Unstructured data with no possible use in ML D. Time-series data

Correct Answer: B

Explanation: Because every ticket already has a known, correct category attached to it (the tag), this is labeled data — the exact kind of data supervised learning requires.

Why the others are wrong: A. Unlabeled data has no predefined answers attached; this data clearly does. C. The text may be unstructured, but that doesn't make it unusable for ML, and the question is about labeling, not structure. D. Time-series data is sequential data collected over time; nothing in the scenario indicates a time dimension.

Task statement mapping: Task 1.1 — Explain basic AI concepts and terminologies. Key concept tested: Labeled vs. unlabeled data.

Question 4 (Beginner) Which AWS service would a company use to automatically convert a written product manual into natural-sounding spoken audio for an accessibility feature?

A. Amazon Transcribe B. Amazon Comprehend C. Amazon Polly D. Amazon Translate

Correct Answer: C

Explanation: Amazon Polly is the AWS text-to-speech service, designed specifically to convert written text into natural-sounding spoken audio.

Why the others are wrong: A. Transcribe performs the opposite conversion: speech to text. B. Comprehend analyzes the meaning of text; it does not produce audio. D. Translate converts text between languages; it does not generate speech.

Task statement mapping: Task 1.2 — Identify practical use cases for AI. Key concept tested: AWS managed AI service identification.

Question 5 (Beginner) Which type of machine learning would be most appropriate for a retailer that wants to discover natural groupings among its customers, given that no predefined customer categories currently exist?

A. Supervised learning B. Unsupervised learning C. Reinforcement learning D. Semi-supervised learning

Correct Answer: B

Explanation: Because no labels or predefined categories exist, and the goal is to discover hidden structure within the data, unsupervised learning (specifically clustering) is the correct approach.

Why the others are wrong: A. Supervised learning requires labeled data with known categories, which the retailer does not have. C. Reinforcement learning requires an agent taking actions in an environment with rewards, which doesn't match this scenario. D. Semi-supervised learning requires at least some labeled data, which is not described here.

Task statement mapping: Task 1.1 — Explain basic AI concepts and terminologies. Key concept tested: Selecting the correct learning type for a scenario.

Question 6 (Beginner) A company wants to process a large archive of scanned, handwritten insurance claim forms to extract the policy number and claimant name from each form into a database. Which AWS service is the best fit?

A. Amazon Rekognition B. Amazon Textract C. Amazon Comprehend D. Amazon Polly

Correct Answer: B

Explanation: Amazon Textract is purpose-built to extract printed and handwritten text, along with structured form fields, from scanned documents and images.

Why the others are wrong: A. Rekognition analyzes visual content like objects and faces, not written text within documents. C. Comprehend analyzes the meaning of text that is already in digital form; it cannot extract text from an image. D. Polly converts text to speech, which is unrelated to extracting data from scanned forms.

Task statement mapping: Task 1.2 — Identify practical use cases for AI. Key concept tested: AWS managed AI service identification.

Question 7 (Beginner) What does the term “inference” mean in the context of machine learning?

A. The process of cleaning and preparing raw data before training B. The process of a trained model making a prediction on new data C. The process of labeling a dataset for supervised learning D. The process of selecting which algorithm to use for a problem

Correct Answer: B

Explanation: Inference refers to the stage where an already-trained model is used to generate predictions on new, real-world input data.

Why the others are wrong: A. This describes data preprocessing, an earlier lifecycle stage. C. This describes a labeling activity, unrelated to inference. D. This describes model selection, which happens before training, not after.

Task statement mapping: Task 1.1 — Explain basic AI concepts and terminologies. Key concept tested: Training vs. inference.

Question 8 (Beginner) A bank needs to process a large batch of overnight mortgage applications and does not need results instantly — results are needed by the time staff arrive the next morning. Which type of inferencing is most appropriate?

A. Real-time inferencing B. Asynchronous inferencing C. Batch inferencing D. Serverless inferencing

Correct Answer: C

Explanation: Batch inferencing is designed for processing large volumes of data together on a schedule when immediate, individual results are not required — exactly matching the overnight processing scenario described.

Why the others are wrong: A. Real-time inferencing is for cases needing an immediate, individual response, which isn’t required here. B. Asynchronous inferencing handles individual large requests with delayed responses, but is not specifically designed for scheduled bulk processing the way

batch is. D. Serverless inferencing refers to a deployment model that scales automatically without managing servers, not specifically to the timing of large overnight jobs.

Task statement mapping: Task 1.1 — Explain basic AI concepts and terminologies. Key concept tested: Types of inferencing.

Question 9 (Beginner) Which term describes a configuration setting, such as the number of training iterations, that is chosen before training begins rather than learned by the model itself?

A. Feature B. Label C. Hyperparameter D. Ground truth

Correct Answer: C

Explanation: A hyperparameter is a setting configured before training starts that controls how the learning process behaves; it is not learned from the data the way model parameters are.

Why the others are wrong: A. A feature is an input variable used by the model, not a training configuration setting. B. A label is the known correct answer attached to training data, unrelated to configuration settings. D. Ground truth refers to the verified, real-world correct answer used for evaluation, not a training setting.

Task statement mapping: Task 1.1 — Explain basic AI concepts and terminologies. Key concept tested: Core ML vocabulary.

Question 10 (Beginner) A company wants to automatically detect the sentiment (positive, negative, or neutral) expressed in thousands of customer product reviews. Which AWS service is the best fit?

A. Amazon Translate B. Amazon Comprehend C. Amazon Textract D. Amazon Polly

Correct Answer: B

Explanation: Amazon Comprehend is purpose-built for analyzing text to detect sentiment, entities, and key phrases, making it the correct fit for sentiment analysis at scale.

Why the others are wrong: A. Translate changes the language of text; it does not analyze sentiment. C. Textract extracts text from scanned images and documents; the reviews are already digital text, so extraction isn't needed. D. Polly converts text into speech and has no sentiment analysis capability.

Task statement mapping: Task 1.2 — Identify practical use cases for AI. Key concept tested: AWS managed AI service identification.

Intermediate Questions (11–20)

Question 11 (Intermediate) A model achieves 98% accuracy on a fraud detection dataset where only 1% of transactions are actually fraudulent. What is the most likely explanation for this result?

A. The model has perfectly solved fraud detection and should be deployed immediately. B. The model may simply be predicting “not fraud” for almost every transaction, making accuracy a

misleading metric here. C. The model has definitely overfit the training data. D. The high accuracy proves the model has high recall as well.

Correct Answer: B

Explanation: When one class (fraud) is rare, a model can achieve very high accuracy simply by predicting the majority class almost every time, while still failing to catch the rare but important fraud cases — which is exactly why precision and recall matter more than accuracy in imbalanced scenarios.

Why the others are wrong: A. High accuracy alone does not confirm the model is actually useful for catching fraud; further metrics like recall are needed. C. Overfitting is a possibility but isn't confirmed by accuracy alone — the scenario describes a class imbalance issue, not necessarily overfitting. D. Accuracy says nothing about recall directly; a model could have high accuracy and very low recall on the rare class.

Task statement mapping: Task 1.3 — Describe the AI/ML development lifecycle. Key concept tested: Limitations of accuracy as a metric.

Question 12 (Intermediate) A hospital is building a model to predict patient readmission risk and must be able to explain to regulators exactly why each individual prediction was made. Which approach is most appropriate?

A. A foundation model accessed through Amazon Bedrock, because foundation models are always more accurate. B. A traditional ML model with strong explainability characteristics, built using Amazon SageMaker AI. C. A generative AI chatbot that explains its own reasoning in natural language. D. Any model will work equally well, since regulators only care about accuracy.

Correct Answer: B

Explanation: When strict explainability and regulatory transparency are required, traditional ML models — which tend to be simpler and easier to interpret — are generally favored over large, complex foundation models, which can be harder to explain in detail.

Why the others are wrong: A. Foundation models are not inherently more accurate for every use case, and their complexity often makes them harder to explain, not easier. C. A chatbot generating natural-language explanations does not guarantee the explanation reflects the model's actual internal reasoning, and introduces hallucination risk. D. Regulators in many industries explicitly require explainability, not just accuracy, making model choice highly relevant.

Task statement mapping: Task 1.2 — Identify practical use cases for AI. Key concept tested: Traditional ML vs. foundation model selection.

Question 13 (Intermediate) During model evaluation, a data science team notices their model performs extremely well on the training data but performs poorly on the validation data. What is this scenario describing?

A. Underfitting B. Overfitting C. Data drift D. Model drift

Correct Answer: B

Explanation: A large gap between strong training performance and weak validation performance is the classic symptom of overfitting — the model has memorized specifics of the training data rather than learning patterns that generalize.

Why the others are wrong: A. Underfitting would show poor performance on both the training and validation data, not just the validation data. C. Data drift refers to a shift in real-world production data over time, not a training-versus-validation performance gap during development. D. Model drift refers to declining performance of an already-deployed model over time, not an evaluation-stage comparison.

Task statement mapping: Task 1.3 — Describe the AI/ML development lifecycle. Key concept tested: Overfitting recognition.

Question 14 (Intermediate) A company is deciding between hosting its own model on Amazon EC2 instances versus using a fully managed API service. Which factor is the clearest advantage of the managed API approach?

A. It guarantees lower total cost in every scenario. B. The cloud provider handles the underlying infrastructure, reducing operational burden on the company. C. It eliminates the need for the company to choose a model at all. D. It is the only option that supports real-time inference.

Correct Answer: B

Explanation: The defining advantage of a managed API service is that the cloud provider operates and maintains the underlying infrastructure, letting the customer focus on using the model rather than managing servers, patches, and scaling.

Why the others are wrong: A. Managed services are not always cheaper in every case; cost depends on usage patterns and scale. C. The company still must choose which model or service to use; a managed API doesn't remove that decision. D. Self-hosted infrastructure on EC2 can also support real-time inference; this is not a unique advantage of managed APIs.

Task statement mapping: Task 1.3 — Describe the AI/ML development lifecycle. Key concept tested: Managed API vs. self-hosted deployment.

Question 15 (Intermediate) A logistics company wants to predict the exact number of packages it will need to ship next month based on historical shipping volume. Which ML technique is most appropriate?

A. Classification B. Clustering C. Regression D. Reinforcement learning

Correct Answer: C

Explanation: Predicting an exact numerical quantity (the number of packages) is a continuous-value prediction problem, which is the definition of regression.

Why the others are wrong: A. Classification predicts a category, not a continuous number. B. Clustering groups similar data points together; it does not predict future numerical values. D. Reinforcement learning is based on agents, actions, and rewards, which doesn't fit this forecasting scenario.

Task statement mapping: Task 1.2 — Identify practical use cases for AI. Key concept tested: Selecting the correct ML technique.

Question 16 (Intermediate) Select TWO. Which of the following are true characteristics of unsupervised learning?

A. It requires a fully labeled dataset before training can begin. B. It can be used to detect anomalies without any predefined examples of what an anomaly looks like. C. It always produces a single predicted numerical value as output. D. It can discover natural groupings within data without predefined categories. E. It is the same technique used in reinforcement learning.

Correct Answer: B, D

Explanation: Unsupervised learning is defined by its ability to work without labels, discovering structure such as natural groupings (clustering) and identifying unusual data points (anomaly detection) purely from patterns in the data itself.

Why the others are wrong: A. This describes supervised learning, the opposite of unsupervised learning. C. Predicting a single numerical value describes regression, a supervised learning technique, not unsupervised learning. E. Reinforcement learning is a distinct learning type based on agents, actions, and rewards — it is not the same as unsupervised learning.

Task statement mapping: Task 1.1 — Explain basic AI concepts and terminologies. Key concept tested: Characteristics of unsupervised learning.

Question 17 (Intermediate) A media company needs to generate searchable, time-stamped transcripts from a large library of recorded webinars. Which AWS service should the company use?

A. Amazon Translate B. Amazon Transcribe C. Amazon Polly D. Amazon Lex

Correct Answer: B

Explanation: Amazon Transcribe is purpose-built to convert recorded speech into accurate, time-stamped text transcripts, exactly matching this use case.

Why the others are wrong: A. Translate converts text between languages; the webinars are audio, not text needing translation. C. Polly converts text into speech, the opposite of what's needed here. D. Lex is for building conversational chatbot interfaces, not for transcription.

Task statement mapping: Task 1.2 — Identify practical use cases for AI. Key concept tested: AWS managed AI service identification.

Question 18 (Intermediate) What is the primary purpose of a validation dataset during model development?

A. To provide the final, unbiased measure of model performance before deployment B. To teach the model the underlying patterns in the data for the first time C. To tune the model and compare different versions during development D. To serve as the live, real-world data the model encounters after deployment

Correct Answer: C

Explanation: The validation set is used throughout development to tune hyperparameters and compare different model versions, helping the team choose the best-performing configuration before final testing.

Why the others are wrong: A. This describes the role of the test set, not the validation set. B. This describes the role of the training set, not the validation set. D. This describes live production data, which is different from a held-out validation set used during development.

Task statement mapping: Task 1.3 — Describe the AI/ML development lifecycle. Key concept tested: Training vs. validation vs. test sets.

Question 19 (Intermediate) Six months after deployment, a retail demand-forecasting model's predictions have become noticeably less accurate, even though the model's code has not changed. The team determines that customer buying patterns have shifted significantly since the model was trained. What should the team most likely do next?

A. Increase the model's accuracy metric threshold to compensate. B. Retrain the model using more recent data to reflect the new buying patterns. C. Switch from batch inferencing to real-time inferencing to fix the issue. D. Conclude that the model was overfit from the very beginning.

Correct Answer: B

Explanation: A decline in performance caused by real-world data shifting away from the original training data is a textbook example of data drift, and the standard response is to retrain the model on fresh, representative data.

Why the others are wrong: A. Adjusting an accuracy threshold doesn't fix the underlying problem of a stale model; it just changes how the problem is measured. C. Changing the inference type (batch vs. real-time) addresses speed and architecture, not the accuracy decline caused by drift. D. The scenario describes a change occurring after deployment due to shifting customer behavior, not an original training flaw like overfitting.

Task statement mapping: Task 1.3 — Describe the AI/ML development lifecycle. Key concept tested: Data drift and retraining.

Question 20 (Intermediate) A company is building a custom fraud detection model using its own proprietary transaction data and wants full control over the algorithm, training process, and deployment infrastructure. Which AWS service is the best fit?

A. Amazon Comprehend B. Amazon Bedrock C. Amazon SageMaker AI D. Amazon Personalize

Correct Answer: C

Explanation: Amazon SageMaker AI is the comprehensive platform for building, training, and deploying fully custom machine learning models, giving the company the control over data, algorithms, and infrastructure it requires.

Why the others are wrong: A. Comprehend is a pre-built NLP service for analyzing text; it doesn't allow training a fully custom model on proprietary structured transaction data. B. Bedrock provides access to existing foundation models through an API; it isn't designed for training a custom model

from scratch on tabular transaction data. D. Personalize is a pre-built recommendation engine service, unrelated to custom fraud detection modeling.

Task statement mapping: Task 1.2 — Identify practical use cases for AI. Key concept tested: SageMaker AI vs. other AWS AI services.

Exam-Ready Questions (21–25)

Question 21 (Exam-Ready) A healthcare technology company is choosing between a traditional ML model and a foundation model for a new clinical decision-support tool. Regulatory requirements demand that every prediction be traceable to specific, interpretable factors, and the company has a relatively small, well-labeled internal dataset of past patient outcomes. Which option best fits these constraints?

A. A foundation model fine-tuned through Amazon Bedrock, since fine-tuning always produces fully explainable outputs. B. A traditional ML model trained on the internal dataset using Amazon SageMaker AI, prioritizing explainability and operating within the available labeled data. C. An off-the-shelf generative AI chatbot, since it can describe its reasoning in plain language for regulators. D. An unsupervised clustering model, since clustering naturally satisfies explainability requirements.

Correct Answer: B

Explanation: Given a small, well-labeled dataset and a hard requirement for traceable, interpretable predictions, a traditional ML model trained on SageMaker AI is the better fit — traditional models are generally easier to interpret than large foundation models, and the company already has the labeled data supervised learning needs.

Why the others are wrong: A. Fine-tuning a foundation model does not guarantee full explainability; foundation models remain harder to interpret in detail than simpler traditional models. C. A chatbot’s self-described reasoning is not a reliable substitute for genuine model interpretability and risks hallucinated explanations. D. Clustering is an unsupervised technique for discovering groupings; it does not produce the kind of individual, traceable predictions a clinical decision-support tool needs, and it doesn’t use the labeled data the company already has.

Task statement mapping: Task 1.2 — Identify practical use cases for AI. Key concept tested: Traditional ML vs. foundation model selection under regulatory constraints.

Question 22 (Exam-Ready) Select TWO. A company is designing the data pipeline for a new credit risk model and wants to follow ML lifecycle best practices. Which two actions reflect appropriate stages of the AI/ML development lifecycle?

A. Skipping data preprocessing because the raw data “looks fine” upon a quick visual check. B. Using a held-out test set, never previously used during development, for the final performance evaluation. C. Deploying the model and considering the project complete with no further monitoring. D. Continuously monitoring the deployed model for data drift and retraining when performance declines. E. Selecting features at random without any business or data analysis.

Correct Answer: B, D

Explanation: Using a fully held-out test set for final evaluation ensures an unbiased performance measurement, and continuously monitoring a deployed model for drift with retraining as needed reflects the ongoing nature of the ML lifecycle's final stage.

Why the others are wrong: A. Skipping preprocessing based on a quick visual check risks missing data quality issues that can seriously harm model performance. C. Treating deployment as the final step ignores the critical monitoring and maintenance stage of the lifecycle. E. Selecting features without business or data analysis undermines feature engineering and likely produces a weaker model.

Task statement mapping: Task 1.3 — Describe the AI/ML development lifecycle. Key concept tested: End-to-end lifecycle best practices.

Question 23 (Exam-Ready) A startup's customer service team wants a system that can both understand spoken customer requests over the phone and respond with natural-sounding spoken answers, while also intelligently routing the conversation based on customer intent. Which combination of AWS services best supports this requirement?

A. Amazon Translate alone, since it processes all forms of language. B. Amazon Lex for conversational intent and dialogue management, combined with Amazon Polly for generating spoken responses. C. Amazon Rekognition for understanding speech, combined with Amazon Textract for generating responses. D. Amazon SageMaker AI alone, since it can be configured to do anything with enough custom training.

Correct Answer: B

Explanation: Amazon Lex is purpose-built to manage conversational intent and dialogue flow, while Amazon Polly generates the natural-sounding spoken responses — together they directly support a voice-based, intent-driven customer service experience.

Why the others are wrong: A. Translate converts between written languages; it does not manage conversation flow or generate speech. C. Rekognition analyzes images and video, not speech, and Textract extracts text from documents, not generates spoken responses — neither fits this audio conversation use case. D. While SageMaker AI is highly flexible, building this entire capability from scratch would ignore purpose-built managed services that already solve this exact problem more efficiently.

Task statement mapping: Task 1.2 — Identify practical use cases for AI. Key concept tested: Combining AWS managed AI services for a real-world solution.

Question 24 (Exam-Ready) A data science team is evaluating a new model intended to detect a rare but serious medical condition that occurs in only 2% of patients screened. Missing an actual case (a false negative) is considered far more dangerous than incorrectly flagging a healthy patient for further testing (a false positive). Which metric should the team prioritize when selecting the best model?

A. Overall accuracy, since it reflects performance across the entire dataset B. Precision, since it measures how many flagged cases were truly positive C. Recall, since it measures how many actual cases were successfully identified D. F1 score only, ignoring precision and recall individually

Correct Answer: C

Explanation: Because missing an actual positive case (a false negative) carries the most serious consequence in this scenario, the team should prioritize recall, which specifically measures the percentage of actual positive cases the model successfully identifies.

Why the others are wrong: A. With only 2% of patients having the condition, accuracy can be extremely high while still missing most or all actual cases, making it a misleading metric here. B. Precision focuses on minimizing false positives, but in this scenario false negatives are the greater concern, not false positives. D. While F1 score balances precision and recall, this scenario specifically calls for prioritizing recall over a balanced metric, since false negatives carry disproportionately higher risk.

Task statement mapping: Task 1.3 — Describe the AI/ML development lifecycle. Key concept tested: Choosing the right evaluation metric for a business scenario.

Question 25 (Exam-Ready) A manufacturing company has a large amount of unlabeled sensor data from its factory equipment and wants to identify unusual equipment behavior that might indicate an upcoming failure, without having historical examples of past failures to learn from. Which approach is most appropriate?

A. Supervised classification trained on labeled failure examples, since this is the only valid approach to predictive maintenance B. Unsupervised anomaly detection, identifying sensor readings that deviate significantly from normal operating patterns C. Reinforcement learning, since the equipment can be treated as an agent receiving rewards D. Use Amazon Polly to generate audio alerts whenever any sensor produces a new value

Correct Answer: B

Explanation: With no labeled examples of past failures available, unsupervised anomaly detection is the appropriate approach, since it identifies unusual patterns directly from the data without requiring predefined examples of what a “failure” looks like.

Why the others are wrong: A. Supervised classification requires labeled examples of past failures, which the company explicitly does not have. C. Reinforcement learning requires an agent taking actions and receiving rewards in an environment; passively monitoring sensor data doesn’t fit this framework. D. Polly only converts text into speech and has no capability to analyze sensor data or detect anomalies.

Task statement mapping: Task 1.2 — Identify practical use cases for AI. Key concept tested: Selecting unsupervised learning for anomaly detection without labeled data.

PART 7: 30 DOMAIN 1 FLASHCARDS

Card #: 1

Domain: 1 - Fundamentals of AI and ML

Category: AI Term

Difficulty: Beginner

Front: What is Artificial Intelligence (AI)?

Back: The broad field of building machines that can perform tasks normally requiring human-like judgment, such as

Card #: 2

Domain: 1 - Fundamentals of AI and ML

Category: AI Term

Difficulty: Beginner

Front: What is Machine Learning (ML)?

Back: A subset of AI where systems learn patterns from data instead of following fixed, hand-coded rules. Every

Card #: 3

Domain: 1 - Fundamentals of AI and ML

Category: AI Term

Difficulty: Beginner

Front: What is Deep Learning?

Back: A subset of machine learning that uses neural networks with many stacked layers to find complex patterns

Card #: 4

Domain: 1 - Fundamentals of AI and ML

Category: ML Concept

Difficulty: Beginner

Front: What is the difference between training and inference?

Back: Training is the process of feeding data into an algorithm so it learns patterns and produces a model. Infer

Card #: 5

Domain: 1 - Fundamentals of AI and ML

Category: ML Concept

Difficulty: Beginner

Front: What is labeled data?

Back: Data where each example already has the correct answer (a “label”) attached, such as a transaction tagged

Card #: 6

Domain: 1 - Fundamentals of AI and ML

Category: ML Concept

Difficulty: Intermediate

Front: What is overfitting?

Back: When a model learns the training data too precisely, including its noise and quirks, causing it to perform v

Card #: 7

Domain: 1 - Fundamentals of AI and ML

Category: ML Concept
Difficulty: Intermediate
Front: What is underfitting?
Back: When a model is too simple to capture the real underlying pattern in the data, leading to poor performance

Card #: 8
Domain: 1 - Fundamentals of AI and ML
Category: ML Concept
Difficulty: Intermediate
Front: What is the difference between bias and variance in ML?
Back: Bias is a systematic error in predictions, often from unrepresentative training data. Variance is how much

Card #: 9
Domain: 1 - Fundamentals of AI and ML
Category: ML Concept
Difficulty: Exam-Ready
Front: Why can accuracy be a misleading metric?
Back: When one class is rare (such as fraud, at 1% of transactions), a model can score very high accuracy just by

Card #: 10
Domain: 1 - Fundamentals of AI and ML
Category: ML Concept
Difficulty: Exam-Ready
Front: What is the difference between data drift and model drift?
Back: Data drift is when real-world input data shifts away from the training data statistically. Model drift is wh

Card #: 11
Domain: 1 - Fundamentals of AI and ML
Category: AWS Service
Difficulty: Beginner
Front: What does Amazon Rekognition do?
Back: A computer vision service that analyzes images and video to detect objects, faces, text, scenes, and inappr

Card #: 12
Domain: 1 - Fundamentals of AI and ML
Category: AWS Service
Difficulty: Beginner
Front: What does Amazon Comprehend do?
Back: A natural language processing service that detects sentiment, identifies entities, classifies topics, and extra

Card #: 13
Domain: 1 - Fundamentals of AI and ML
Category: AWS Service
Difficulty: Beginner
Front: What does Amazon Transcribe do?
Back: An automatic speech recognition (ASR) service that converts spoken audio into accurate written text transcriptions

Card #: 14
Domain: 1 - Fundamentals of AI and ML
Category: AWS Service
Difficulty: Beginner
Front: What does Amazon Polly do?
Back: A text-to-speech service that converts written text into natural-sounding spoken audio, supporting many languages

Card #: 15
Domain: 1 - Fundamentals of AI and ML
Category: AWS Service
Difficulty: Beginner
Front: What does Amazon Translate do?
Back: A neural machine translation service that converts text from one language into another, used for content localization

Card #: 16
Domain: 1 - Fundamentals of AI and ML
Category: AWS Service
Difficulty: Intermediate
Front: What does Amazon Textract do, and how is it different from Rekognition?
Back: Textract extracts printed and handwritten text and structured form data from scanned documents and images

Card #: 17
Domain: 1 - Fundamentals of AI and ML
Category: AWS Service
Difficulty: Intermediate
Front: What does Amazon Lex do?
Back: A service for building conversational interfaces — chatbots and voice bots — that understand natural language

Card #: 18
Domain: 1 - Fundamentals of AI and ML
Category: AWS Service
Difficulty: Intermediate
Front: What does Amazon SageMaker Ground Truth do?
Back: A SageMaker AI feature that helps create high-quality labeled datasets using human labelers combined with machine learning

Card #: 19

Domain: 1 - Fundamentals of AI and ML

Category: Comparison

Difficulty: Beginner

Front: What is the difference between Amazon Polly and Amazon Transcribe?

Back: They are opposites. Polly converts text into speech (text-to-speech). Transcribe converts speech into text

Card #: 20

Domain: 1 - Fundamentals of AI and ML

Category: Comparison

Difficulty: Beginner

Front: What is the difference between classification and regression?

Back: Classification predicts a category or class (such as “spam” vs. “not spam”). Regression predicts a continuous

Card #: 21

Domain: 1 - Fundamentals of AI and ML

Category: Comparison

Difficulty: Intermediate

Front: What is the difference between Amazon SageMaker AI and Amazon Bedrock?

Back: SageMaker AI is for building, training, and deploying your own custom ML models from scratch. Bedrock

Card #: 22

Domain: 1 - Fundamentals of AI and ML

Category: Comparison

Difficulty: Intermediate

Front: What is the difference between Amazon Comprehend and Amazon Textract?

Back: Textract extracts text from images and scanned documents into machine-readable form. Comprehend anal

Card #: 23

Domain: 1 - Fundamentals of AI and ML

Category: Comparison

Difficulty: Intermediate

Front: What is the difference between precision and recall?

Back: Precision measures, of everything flagged as positive, how many were truly positive (minimizing false posit

Card #: 24

Domain: 1 - Fundamentals of AI and ML

Category: Comparison

Difficulty: Intermediate

Front: What is the difference between the validation set and the test set?

Back: The validation set is used during development to tune the model and compare versions, often repeatedly.

Card #: 25

Domain: 1 - Fundamentals of AI and ML

Category: Comparison

Difficulty: Exam-Ready

Front: What is the difference between batch inferencing and real-time inferencing?

Back: Real-time inferencing delivers immediate, individual predictions (low latency, higher cost). Batch inferencing

Card #: 26

Domain: 1 - Fundamentals of AI and ML

Category: Scenario

Difficulty: Beginner

Front: A company has thousands of customer records with no predefined categories and wants to discover natural

Back: Unsupervised learning (specifically clustering) — it discovers hidden groupings in unlabeled data without

Card #: 27

Domain: 1 - Fundamentals of AI and ML

Category: Scenario

Difficulty: Intermediate

Front: A hospital needs a model whose predictions can be fully explained to regulators, and has a small, well-labeled

Back: Traditional ML model, built on Amazon SageMaker AI. Traditional models are generally easier to explain

Card #: 28

Domain: 1 - Fundamentals of AI and ML

Category: Scenario

Difficulty: Intermediate

Front: A bank wants to process a large batch of mortgage applications overnight with no need for instant results

Back: Batch inferencing — it is designed for processing large volumes of data together on a schedule, which matches

Card #: 29

Domain: 1 - Fundamentals of AI and ML

Category: Scenario

Difficulty: Exam-Ready

Front: A factory has unlabeled sensor data and no historical failure examples but wants to flag unusual equipment

Back: Unsupervised anomaly detection — it identifies data points that deviate from normal patterns without requiring

Card #: 30

Domain: 1 - Fundamentals of AI and ML

Category: Scenario

Difficulty: Exam-Ready

Front: A medical screening model must minimize missed diagnoses (false negatives) even if it means more false a

Back: Recall — it measures how many actual positive cases the model successfully identifies, which is the priority

PART 8: DOMAIN 1 ONE-PAGE CHEAT SHEET

Top 10 Concepts to Understand (Not Memorize)

1. AI contains ML; ML contains deep learning; deep learning contains GenAI — a nested hierarchy, not separate categories.
2. Supervised learning needs labels; unsupervised learning does not; reinforcement learning uses rewards instead of labels.
3. Classification predicts a category; regression predicts a number; clustering discovers groups.
4. The ML lifecycle has 8 stages, from problem identification through ongoing monitoring — it never truly “ends” at deployment.
5. Overfitting = memorizing training data; underfitting = too simple to learn the pattern at all.
6. Accuracy can lie when classes are imbalanced — precision and recall reveal the real story.
7. Recall matters most when missing a positive case is dangerous (fraud, disease); precision matters most when false alarms are costly.
8. Batch inferencing = scheduled, bulk, cheaper, slower. Real-time inferencing = instant, individual, costlier, faster.
9. Traditional ML models tend to be more explainable; foundation models tend to be more flexible but harder to explain.
10. Data drift = the world changed; model drift = the model’s accuracy declined as a result. Both trigger retraining.

AWS Services at a Glance

Service	Exam-Ready Description
Amazon SageMaker AI	Build, train, and deploy your own custom ML models
Amazon Rekognition	Analyze images and video for objects, faces, and scenes
Amazon Comprehend	Analyze text for sentiment, entities, and topics
Amazon Transcribe	Convert speech into text
Amazon Translate	Convert text between languages
Amazon Polly	Convert text into speech
Amazon Textract	Extract text and form data from scanned documents
Amazon Lex	Build chatbots and voice bots
Amazon Personalize	Build personalized recommendation systems

Essential Vocabulary, One Line Each

- **Algorithm:** the method or steps used to learn from data.
- **Model:** the trained result that makes predictions.
- **Feature:** an input variable used by the model.
- **Label:** the known correct answer in training data.
- **Hyperparameter:** a setting chosen before training begins.
- **Overfitting:** memorized training data, fails on new data.
- **Underfitting:** too simple, fails on everything.
- **Bias:** systematic prediction error.
- **Variance:** inconsistency across different training samples.
- **Precision:** of flagged positives, how many were correct.
- **Recall:** of actual positives, how many were caught.

The ML Lifecycle in 8 Words

Identify → Collect → Prepare → Engineer → Train → Evaluate → Deploy → Monitor

5 Exam Watch-Outs for Domain 1

1. Don't confuse **Amazon Polly** (text-to-speech) with **Amazon Transcribe** (speech-to-text) — they are opposites.
2. Don't confuse **Amazon Rekognition** (visual content) with **Amazon Textract** (text extracted from document images).
3. Don't confuse **Amazon SageMaker AI** (build your own model) with **Amazon Bedrock** (use an existing foundation model).
4. Don't confuse **overfitting** (too complex, memorizes) with **underfitting** (too simple, learns nothing meaningful).
5. Don't confuse **data drift** (the input data changed) with **model drift** (the input-output relationship changed, hurting accuracy).

5 Exam Clue Phrase Pairs

1. “Convert text into speech” → **Amazon Polly**.
2. “Convert speech into text” → **Amazon Transcribe**.
3. “No predefined categories, find hidden groups” → **Unsupervised learning / clustering**.
4. “Missing a true case is dangerous” → **Prioritize recall**.
5. “Needs full explainability for regulators” → **Traditional ML model, not a foundation model**.

PART 9: DOMAIN 1 SUMMARY AND REVIEW PLAN

5 Key Takeaways from Domain 1

1. AI, ML, deep learning, and GenAI form a nested hierarchy — not four separate, competing technologies. Understanding containment (what fits inside what) resolves a large share of Domain 1 questions instantly.

2. The four types of learning (supervised, unsupervised, semi-supervised, reinforcement) are distinguished primarily by what kind of data is available — fully labeled, fully unlabeled, partially labeled, or no labels at all (just rewards).
3. The ML development lifecycle has eight connected stages, and the journey doesn't stop at deployment — monitoring and retraining are permanent, ongoing responsibilities.
4. Accuracy alone is often the wrong metric to trust, especially with imbalanced data; precision and recall tell a more complete and business-relevant story.
5. AWS managed AI services each solve one clear, narrow job — memorizing the clue phrase for each service (and its most commonly confused counterpart) is one of the highest-value study activities for this domain.

Remediation Plan: If You Score Below 60% on Domain 1 Practice Questions

1. **Start with Part 3A (the AI hierarchy).** Most low scores trace back to confusion between AI, ML, deep learning, and GenAI. Re-read this section and redraw the nested diagram from memory.
2. **Move to Part 3B (types of ML) and Table 2.** Confirm you can state, without hesitation, what kind of data each learning type needs.
3. **Review Part 3E (AWS services) and the “most commonly confused with” notes for every service.** Service-pairing questions (Polly vs. Transcribe, Rekognition vs. Textextract, SageMaker AI vs. Bedrock) are a recurring theme across the entire exam, not just Domain 1.
4. **Revisit Part 5, Table 7 (accuracy, precision, recall, F1).** If metric-selection questions are a weak spot, focus specifically on the “best used when” column — these scenario cues repeat across the exam.
5. **Retake the 25 practice questions in Part 6,** this time reading every explanation (including for the answers you got right) before moving to Part 4 (Domain 2).

3-Day Focused Domain 1 Review Schedule

Day 1 — Concepts (60–90 minutes): Re-read Part 1 (introduction and misconceptions), Part 3A (hierarchy), Part 3B (learning types), and Part 3C (lifecycle). Self-test by explaining each concept aloud in your own words without looking at the page.

Day 2 — Services and Vocabulary (60–90 minutes): Re-read Part 3D (use cases) and Part 3E (AWS services), focusing especially on the “exam clue phrases” and “most commonly confused with” sections. Then work through the Part 4 term table, covering the “Back” column and quizzing yourself.

Day 3 — Practice (60–90 minutes): Complete all 25 practice questions in Part 6 under light time pressure (aim for under 90 seconds per question). Review every explanation, then run through all 30 flashcards in Part 7 twice.

How Domain 1 Content Reappears in Domains 2 Through 5

- **Domain 2 (Fundamentals of GenAI):** Foundation models and large language models are described as advanced applications of the deep learning concepts introduced in Domain 1's AI hierarchy. The training, inference, and lifecycle vocabulary from Domain 1 reappears as the “FM lifecycle” (pre-training, fine-tuning, evaluation, deployment).
- **Domain 3 (Applications of Foundation Models):** Evaluation metrics like accuracy and the general concept of model evaluation extend directly into FM-specific metrics like ROUGE,

BLEU, and LLM-as-a-judge. Fine-tuning is a direct extension of the “training” concept from Domain 1.

- **Domain 4 (Guidelines for Responsible AI):** Bias and variance, first defined here in a statistical sense, are revisited as the foundation for fairness, demographic bias, and dataset balance discussions.
- **Domain 5 (Security, Compliance, and Governance):** The ML pipeline stages from Domain 1 (data collection, training, deployment, monitoring) are the exact stages where security risks — like data leakage or compromised training data — are introduced and must be defended against.

End of Domain 1 Study Module — Task 3 of 15 Document version: 1.0 / Based on AIF-C01 Exam Guide Version 1.1 (April 30, 2026) Next task: Task 4 — Domain 2 Study Module: Fundamentals of GenAI